

Analysis of the Spontaneous Emission of a Tunable Phase-Shift-Controlled DFB Laser Filter

Mouhamad J. Chawki, Jean Le Bihan, M. Le Ligne, and Abdelkarim Zatni

Abstract—Using a transfer matrix method, the results of analysis of the spontaneous emission spectrum of a three section phase controlled DFB laser are presented for implementing a continuously tunable optical filter. The influence of various parameters—net field gain, coupling coefficient, phase shift and reflection coefficients—on the spectrum behavior are studied theoretically. The main- to subtransmission peak magnitude ratio can be chosen while varying the frequency, which demonstrates the tunability of the three-section phase controlled DFB laser filter. Analysis shows that a three section DFB laser (a passive phase shift control region between two DFB structures) presents a single-mode spontaneous emission spectrum operating over a wide continuous tuning range of less than 1 nm. The results of our work should aid in the design of multisection DFB laser filter.

I. INTRODUCTION

A WAVELENGTH filter is a key component for wavelength division multiplexing (WDM) techniques useful for optical fiber communication links. The purpose is to take advantage of the wide-band characteristics of light to transmit a great number of channels. Moreover, devices that allow to realize functions by a pure optical manner are requested because systems that do not require changing from optical to electrical signals are expected to be more and more developed. Various properties are desired: a constant transmission gain, a constant bandwidth, a wide tuning range, and a high rejection rate. Several types of tunable wavelength filters have been laid on the distributed feedback laser diode filter. This device offers several main advantages when operating just below the threshold condition: a narrow bandwidth, an optical gain, a wavelength tunability, and a high rejection rate.

A number of authors have considered different DFB laser structures as tunable wavelength filters. First, let us mention the $\lambda/4$ DFB laser structure. In this structure, it is found that changing the driving current for spectrum selection has the adverse effect of changing both the optical gain and the bandwidth [1].

In the two-electrode DFB laser structure introduced by Magari *et al.* [2], [3], by changing the two currents the lasing frequency of the laser can be controlled while maintaining a constant optical output power and without

any mode jump. For this reason, a narrow spectrum selection (3.6 GHz) with a wide frequency tunability (33.3 GHz) was achieved while maintaining a constant gain and a constant gain bandwidth.

A phase controlled laser structure (one active section and one phase control section) has been introduced by Numai *et al.* [4]. A tuning range as wide as 43 GHz with 27-dB constant gain has been achieved. Finally, these authors have also introduced a three section phase shift controlled DFB laser filter (two active sections separated by a phase control section) with a wide tuning range [5]. A tuning range as wide as 120 GHz and a 24.5-dB constant gain have been achieved.

Moreover, published research deals with distributed Bragg reflector (DBR) filters (i.e., filters with the active region separate from the grating section). Kazovsky *et al.* [6] have shown that two drawbacks of a conventional DBR device, a multilobe transfer function, and a small gain can be alleviated by reduction of the grating coupling coefficient and the physical length of one or the two sections. This technique can lead to devices having a gain of 35 dB, sidelobe suppression of 32 dB and bandwidth as narrow as 1 GHz. A DBR structure has been described for implementing a continuously tunable optical filter [7]. The filter has only one transmission peak over the entire frequency range. It can be tuned very quickly (in a few nanoseconds) and has the potential of being integrated with the receiver in a photonic circuit.

Recently, the attractive capabilities of the DFB laser as a multifunction device have been investigated. The laser can be used for light emission or optical filtering or photodetection. The operating principle of the DFB laser for photodetection consists in detecting the quasi-Fermi level separation which is induced by the optical injected signal. Recent work has demonstrated FSK transmission up to 250 Mb/s using an identical MQW-DBR laser as both a transmitter and a receiver [8]. In the receiver the MQW-DBR laser acts as a tunable filter, FSK discriminator, and photodetector. This new device was referred as a tunable resonant amplifying photodetector (TRAP).

More recently, a FSK transmission has been reported, using two-electrode DFB lasers for both transmitter and receiver [9]. A 1.5-Gb/s transmission with -20 -dBm sensitivities for 10^{-9} bit error rate has been achieved over 40 km of single-mode optical fiber. The transmitter consisted in a two-electrode DFB laser, the receiver which acted as a filter, a FSK discriminator, and a detector was

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M. J. Chawki and M. Le Ligne are with CNET/LAB/SMR/ESO-BP 40, 22301 Lannion Cedex, France.

J. Le Bihan and A. Zatni are with Laboratoire RESO, Ecole Nationale d'Ingénieurs de Brest, 29287 Brest Cedex, France.

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also a two-electrode DFB laser [10]. So such a new simple multifunction device can replace the complexity of coherent FSK demodulator in future WDM systems.

These last years there has been substantial interest in the study of the properties of DFB laser structures for optical filtering. The spectrum behavior below the threshold has been investigated [11], [12]. Makino and Glinski have obtained some results useful for optimizing second-order dynamic single-mode DFB lasers [13]. These authors have also presented a matrix analysis of the amplified spontaneous emission of DFB semiconductor laser amplifiers [14].

The present paper deals with the analysis of the spontaneous emission of a laser, biased just below the threshold condition, to be used as an optical filter, especially with the phase-shift-controlled DFB structure, which presents a wide tuning range [5]. From a matrix analysis of the DFB laser available for a multisection structure, we study a DFB structure with three sections, a phase control section between two active sections. The aim of the study presented here is to analyze the influence of various parameters on the characteristics of the laser frequency dependent power output with the view of optimizing the design of such structures. The stress is laid upon the interesting capability of tunability of the phase-shift-controlled laser filter with high rejection rate.

II. THEORETICAL ANALYSIS

A. Derivation of the Transfer Matrix For an n Section DFB Structure

In Fig. 1, a multisection DFB structure is shown, divided into n sections, with facet field reflection coefficients r_1 and r_2 at positions z_1 and z_{n+1} , respectively. In each section length L_m ($L_m = z_{m+1} - z_m$), corrugations are considered to be uniform and, according to the coupled-mode theory, the forward wave A and the backward wave B can be expressed in a matrix form as

$$\begin{bmatrix} A(z_{n+1}) \\ B(z_{n+1}) \end{bmatrix} = \mathbf{T} \begin{bmatrix} A(z_1) \\ B(z_1) \end{bmatrix} \quad (1)$$

where $\mathbf{T}$ is the transfer matrix of the whole structure, given by the following expression:

$$\mathbf{T} = \mathbf{T}^n \mathbf{T}^{n-1} \dots \mathbf{T}^1 \quad (2)$$

where m is an index.

The transfer matrix of the m^{th} section $\mathbf{T}^m$ can be written as

$$\mathbf{T}^m = \begin{bmatrix} T_{11}^m & T_{12}^m \\ T_{21}^m & T_{22}^m \end{bmatrix} \quad (3)$$

where the matrix elements are

$$\begin{aligned} T_{11}^m &= (E_m - R_m^2 E_m^{-1}) / (1 - R_m^2) \\ T_{12}^m &= -T_{21}^m = -R_m (E_m - E_m^{-1}) / (1 - R_m^2) \\ T_{22}^m &= -(R_m^2 E_m - E_m^{-1}) / (1 - R_m^2) \end{aligned} \quad (4)$$

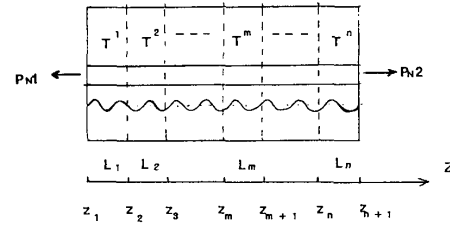


Fig. 1. Schematic diagram of a multisection DFB laser filter.

and

$$E_m = \exp [\gamma_m (z_{m+1} - z_m)] \quad (5)$$

$$R_m = -j\kappa_m / (\gamma_m + \alpha_m + j\delta_m) \quad (6)$$

where γ_m is the propagation constant for the m^{th} section given by

$$\gamma_m^2 = \kappa_m^2 + (\alpha_m + j\delta_m)^2 \quad (7)$$

κ_m and α_m are the coupling coefficient and the net field gain for the m^{th} section, respectively, and δ_m (detuning) is the Bragg parameter indicating the deviation from the Bragg wave number.

Makino and Glinski have theoretically developed the normalized spontaneous emission power (per unit frequency bandwidth) for facet ℓ ($\ell = 1, 2$). It can be expressed by the sum of the contributions from n sections as

$$p_{N\ell} = hfG \sum_{m=1}^n n_{\text{spm}} \chi_{\ell m} \quad (\ell = 1, 2) \quad (8)$$

where the transmission gain G of the whole structure can be written as

$$G = \frac{(1 - |r_1|^2)(1 - |r_2|^2)}{|D|^2} \quad (9)$$

hf is the photon energy, n_{spm} is the ratio of the spontaneous to stimulated emission for the m^{th} section and

$$D = T_{22} + r_1 T_{21} - r_2 T_{12} - r_1 r_2 T_{11}. \quad (10)$$

Here $\chi_{\ell m}$ is the inverse differential quantum efficiency at the threshold [14]. $\chi_{\ell m}$ depends on the above-mentioned different factors: r_1 , r_2 , κ_m , α_m , δ_m and is proportional to g_m , the optical power gain per unit length for the m^{th} section ($g_m = 2\alpha_m + \alpha_{\text{abm}}$, α_{abm} : absorption power loss).

B. Transfer Matrix of a Three Section Phase Shifted DFB Structure

For the structure with a phase shifted region represented in Fig. 2, the transfer matrix is given by

$$\mathbf{T} = \mathbf{T}^3 \mathbf{T}^2 \mathbf{T}^1 \quad (11)$$

where $\mathbf{T}^1$ and $\mathbf{T}^3$ are the DFB region matrices and $\mathbf{T}^2$ corresponds to the phase-shift region matrix whose ele-

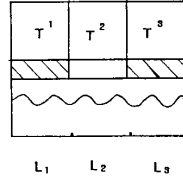


Fig. 2. Schematic diagram of a phase-shifted DFB laser filter.

ments are

$$\mathbf{T}^2 = \begin{bmatrix} \exp(j\psi) & 0 \\ 0 & \exp(-j\psi) \end{bmatrix} \quad (12)$$

where ψ is a phase shift and $R_2 = 0$.

For two identical DFB zones, $L_1 = L_3 = L$, $\kappa_1 = \kappa_3 = \kappa$, $\alpha_1 = \alpha_3 = \alpha$, $\delta_1 = \delta_3 = \delta$, $g_1 = g_3 = g$ and $n_{sp1} = n_{sp3} = n_{sp}$.

So, neglecting the spontaneous emission in the L_2 region [14], the normalized spontaneous emission power for facet 1 can be written as

$$p_{N1} = hfGn_{sp}(\chi_{11} + \chi_{13}). \quad (13)$$

Expressions of χ_{11} and χ_{13} can be shown to be [14]

$$\begin{aligned} \chi_{1m} = & \frac{g}{1 - |r_2|^2} \left\{ (1 + |R|^2) \left[|c_{1m}|^2 \frac{\exp(2\gamma_r L) - 1}{2\gamma_r} \right. \right. \\ & + |d_{1m}|^2 \frac{1 - \exp(-2\gamma_r L)}{2\gamma_r} \left. \right] + 2 \operatorname{Re} \\ & \cdot \left[(R + R^*) c_{1m} d_{1m}^* \frac{\exp(2j\gamma_i L) - 1}{2j\gamma_i} \right] \left. \right\} \\ & m = (1, 3) \end{aligned} \quad (14)$$

where $\gamma_r = \operatorname{Re}(\gamma)$ and $\gamma_i = \operatorname{Im}(\gamma)$, and

$$c_{11} = \frac{(\Phi - R^2 \Phi^{-1})(r_2 - R) \exp(\gamma L) + R(\Phi - \Phi^{-1})(1 - r_2 R) \exp(-\gamma L)}{(1 - R^2)^2} \quad (15)$$

$$d_{11} = \frac{R(\Phi^{-1} - \Phi)(r_2 - R) \exp(\gamma L) + (\Phi^{-1} - R^2 \Phi)(1 - r_2 R) \exp(-\gamma L)}{(1 - R^2)^2} \quad (16)$$

$$c_{13} = \frac{r_2 - R}{1 - R^2} \quad (17)$$

$$d_{13} = \frac{1 - r_2 R}{1 - R^2} \quad (18)$$

where

$$\Phi = \exp(j\psi). \quad (19)$$

Expressions of χ_{21} and χ_{23} can also be written in a similar manner. c_{21} and d_{21} have the same expressions as c_{13} and d_{13} , respectively, c_{23} and d_{23} have the same expressions as c_{11} and d_{11} , respectively. The only difference consists in the change of r_2 into r_1 in relations (14) to (18).

III. RESULTS AND DISCUSSION

In this part, the influence of the main parameters on the behavior of the phase-shift-controlled laser filter is described and discussed.

A. Determination of the Threshold Conditions

To analyze the spontaneous emission spectrum of the laser, the threshold conditions have to be determined. For a given value of κL , the normalized power $\bar{p}_N = p_N / (hf n_{sp})$ depends on the net field gain αL . To clarify this dependence, a theoretical relation between αL and κL is developed in the following.

We consider the case of a three section DFB structure with antireflection coated facets ($r_1 = r_2 = 0$) and a passive region introducing a phase shift $\psi = \pi/2$ ($\Phi = j$). The normalized spontaneous emission power $\bar{p}_N$ presents a main peak located at the Bragg wavelength ($\delta L = 0$).

From (6) to (7), αL and γL can be related to κL as follows:

$$\gamma L = \frac{1}{2} \kappa L \left(jR + \frac{1}{jR} \right) \quad (20)$$

$$\alpha L = \frac{1}{2} \kappa L \left(\frac{1}{jR} - jR \right). \quad (21)$$

The threshold conditions of the DFB laser are achieved when the quantity D becomes null, which according to (10) corresponds to the following relation between R and $E = \exp(\gamma L)$

$$R^2(E - E^{-1})^2 + (R^2 E - E^{-1})^2 = 0 \quad (22)$$

which can be written

$$E^2 = \frac{1 \pm jR}{jR(1 \pm jR)}. \quad (23)$$

For a given value of the coupling coefficient κL , $\bar{p}_N$ depends on the net field gain αL . By considering κL ($\kappa L > 0$), γL ($\gamma L > 0$), αL ($\alpha L > 0$), and jR ($|R| < 1$) as real parameters in the present case, jR is found, according to (21) and (24), by solving the equation

$$\frac{jR}{1 + (jR)^2} \operatorname{Ln} \frac{1 + jR}{jR(1 - jR)} = \kappa L. \quad (24)$$

Then $\gamma_{th}L$ and $\alpha_{th}L$ can be deduced from (21) and (22) for any given value of κL . For example, these numerical values of the parameters

$$\kappa L = 0.5 \quad jR = 0.298 \quad \alpha_{th}L = 0.763 \quad \gamma_{th}L = 0.912$$

$$\kappa L = 1 \quad jR = 0.710 \quad \alpha_{th}L = 0.349 \quad \gamma_{th}L = 1.059$$

$$\kappa L = 1.5 \quad jR = 0.897 \quad \alpha_{th}L = 0.164 \quad \gamma_{th}L = 1.509$$

correspond to threshold conditions.

Fig. 3 shows a plot of $\alpha_{th}L$ versus κL . From this figure, a low threshold net field gain $\alpha_{th}L$ is obtained for high coupling coefficient κL .

B. Effect of the Net Field Gain αL and the Coupling Coefficient κL on the Spectrum

Fig. 4(a) shows the normalized spontaneous emission power $\bar{P}_{N1}$ for facet 1 as a function of the detuning δL , for a first-order grating, with $\alpha L = 0.35$, $\kappa L = 1$, $r_1 = r_2 = 0$ and a phase shift $\psi = \pi/2$. Fig. 4(b) is a plot of the inverse differential quantum efficiency χ versus δL .

It can be seen that the spectrum presents a single peak located at the Bragg wavelength ($\delta L = 0$), instead of the two peaks separated by a stop band in the standard structure, i.e., the monoelectrode DFB laser (Fig. 5), due to the frequency dependence of the χ factor. The spectrum in this case corresponds to that of a $\lambda/4$ shifted DFB laser. The performance of the laser filtering is determined by its rejection rate which depends on the power ratio of the two peaks P_{sub}/P_{main} .

To clarify this dependence, the main power peak $\bar{P}_N$ ($\delta L = 0$) and the sub peaks have their amplitude been plotted in Fig. 6 as functions of αL for $\kappa L = 1$, $r_1 = r_2 = 0$, and $\psi = \pi/2$.

From Fig. 6, it can be noted that the laser spectrum changes with αL . For $\alpha L < 0.5$, the laser spectrum is single mode with a peak located at the Bragg wavelength. For $\alpha L \geq 0.5$, the laser spectrum becomes multimode. In Fig. 7, three spectra relative to three different values of αL are shown with $\kappa L = 1$, $r_1 = r_2 = 0$, and $\psi = \pi/2$. It can be seen that, when the laser filter operates under the threshold conditions (net field gain less than about 0.35), it presents a single-mode spectrum with high power ratio P_{sub}/P_{main} (Fig. 6). This result agrees with the calculation presented in Section III-A. The coupling coefficient κL also influences the laser spectrum which remains single-mode for a suitable value of the net field gain αL .

Fig. 8 shows three spectra for different couples (κL , αL). From this figure, the greatest value of the coupling coefficient κL ($\kappa L = 1.5$) corresponds to the smallest value of the net field gain αL ($\alpha L = 0.169$) (Fig. 3). When κL gets greater (and αL smaller), the mode suppression ratio is improved and the width of the main mode gets narrower. These results have been demonstrated by Magari *et al.* [3].

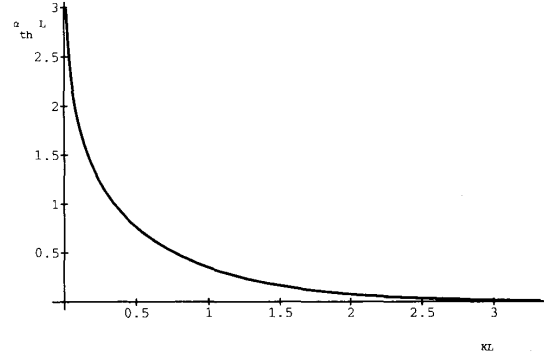


Fig. 3. Threshold net field gain $\alpha_{th}L$ versus coupling coefficient κL .

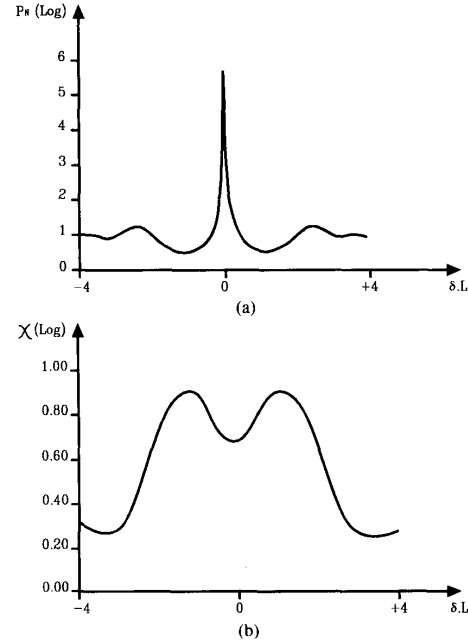


Fig. 4. a) Normalized emission spectrum b) X factor as function of δL for a phase-shifted-controlled DFB laser with nonreflective cavity ends ($\kappa L = 1$, $\alpha L = 0.35$ and $\psi = \pi/2$).

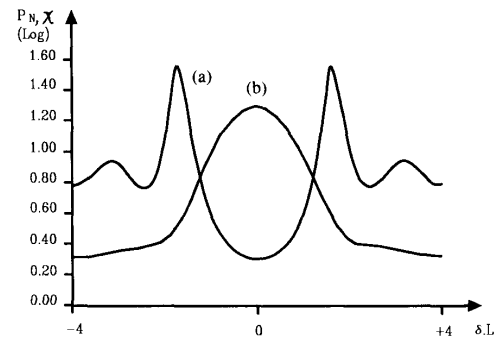


Fig. 5. a) Normalized emission spectra below the threshold b) X factor as function of δL for a standard DFB laser with nonreflective cavity ends ($\kappa L = 1$ and $\alpha L = 0.3$).

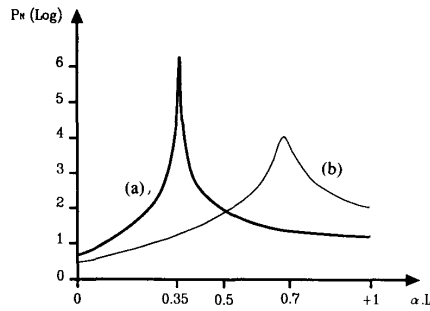


Fig. 6. a) P_{main} and b) P_{sub} versus αL ($KL = 1$, $r_1 = r_2 = 0$, $\psi = \pi/2$).

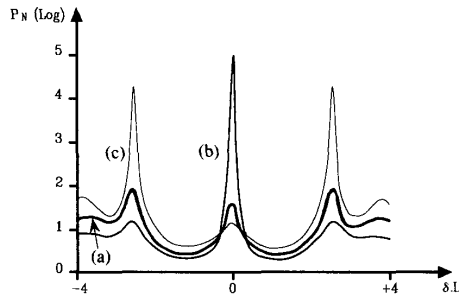


Fig. 7. Normalized emission spectra as function of δL for different values of αL ($KL = 1$, $r_1 = r_2 = 0$ and $\psi = \pi/2$): a) $\alpha L = 0.52$ b) $\alpha L = 0.35$ c) $\alpha L = 0.7$.

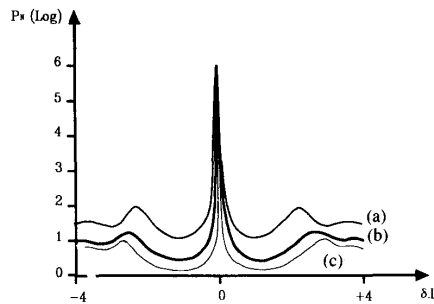


Fig. 8. Normalized emission spectra as function of δL for different couples (KL , αL) ($r_1 = r_2 = 0$ and $\psi = \pi/2$): a) $KL = 0.5$ and $\alpha L = 0.76$ b) $KL = 1$ and $\alpha L = 0.35$ c) $KL = 1.5$ and $\alpha L = 0.169$.

C. Effect of the Phase Shift on the Tuning Range

The laser filter tunability depends on the phase shift introduced by the phase control region. Fig. 9 shows the spectra of the phase-shift-controlled DFB laser filter. The spectra from (a) to (d) are for the laser with various phase shift values for the same αL ($\kappa L = 1.5$, $r_1 = r_2 = 0$). When the phase shift is equal to $\pi/2$, the main power is tuned exactly to the Bragg wavelength. For $\psi = 3\pi/4$ or $\pi/4$ the main peak power presents a red or a blue shift tuning, respectively. This result was predicted in [11].

Fig. 10 shows the magnitude of the main power peak as a function of the phase shift ψ for different values of

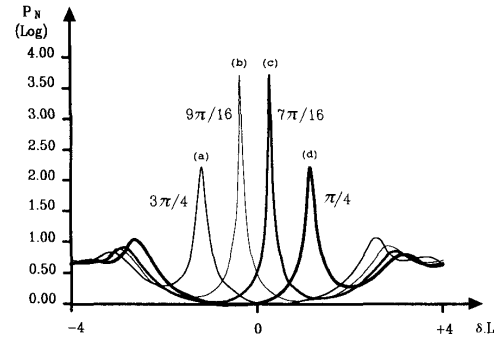


Fig. 9. Normalized emission spectra below the threshold as function of δL with different values of phase shift.

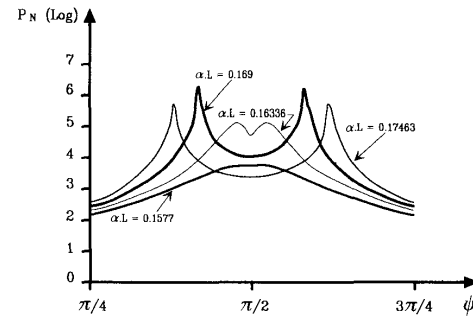


Fig. 10. Relation between the P_{main} and phase shift for different values of αL ($KL = 1.5$ and $r_1 = r_2 = 0$).

the net field gain αL ($\kappa L = 1.5$, $r_1 = r_2 = 0$). For the lower values of αL ($\alpha L < 0.16$), the main power peak presents small change with respect to the phase shift. For the higher values of αL ($\alpha L > 0.16$), two resonance peaks, located symmetrically on either side of the phase shift $\psi = \pi/2$, appear. This is due to the fact that the laser approaches the threshold conditions. Increasing the parameter αL modifies the spectrum behavior (Fig. 6) and these two resonance peaks move apart creating a deep valley between them. The interesting feature of the simulated spectra is that the power ratio of the two peaks $P_{\text{sub}}/P_{\text{main}}$ strongly depends on the phase shift value, especially when the threshold is approached.

Fig. 11 shows the power of the main peak and that of the subpeaks as a function of the phase shift for $\alpha L = 0.1577$ (the laser operates just below the threshold conditions), $\kappa L = 1.5$, $r_1 = r_2 = 0$. It can be noted that the subpeak power presents a rather small variation with respect to ψ .

The shift of the main peak, that characterizes the tunability, as a function of the phase shift ψ is shown in Fig. 12. A tunability of $2 \delta L$ is obtained for a phase shift variation $\Delta\psi$ of $\pi/2$ ($\pi/4 \leq \psi \leq 3\pi/4$).

To estimate the tuning range, we develop a relation between the detuning δL and the variation of the wavelength. The relation between the propagation constant and

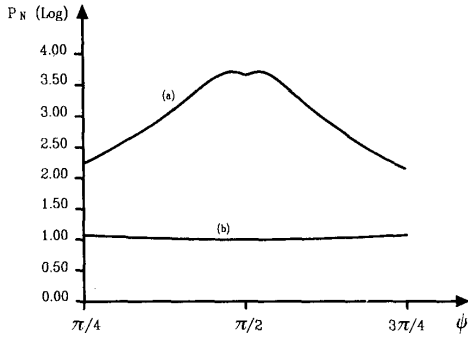


Fig. 11. a) P_{main} and b) P_{sub} versus phase shift for $\alpha L = 0.1577$ ($KL = 1.5$ and r_1 and $r_2 = 0$).

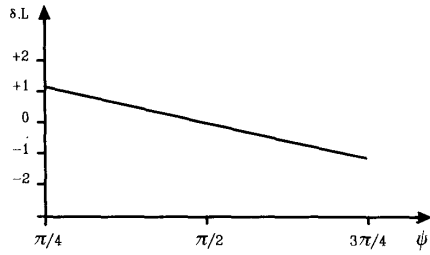


Fig. 12. Tuning of main peak versus phase shift for $\alpha L = 0.1577$ ($KL = 1.5$ and $r_1 = r_2 = 0$).

the wavelength is given by

$$\beta = \frac{2\pi n_g}{\lambda} \quad (25)$$

where n_g is the group index in the semiconductor. If the variation of n_g with respect to λ is neglected, the derivation of β leads to [15]

$$\frac{\Delta\beta}{\Delta\lambda} = -\frac{2\pi n_g}{\lambda^2} \quad (26)$$

So, considering the relation

$$\delta L = \Delta\beta L \quad (27)$$

the tuning range, expressed with respect to δL , is given by

$$\frac{\Delta\lambda}{\delta L} = -\frac{\lambda^2}{2\pi L n_g} \quad (28)$$

Therefore, in an InP semiconductor structure ($n_g \approx 4$), if $\lambda = 1.55 \mu\text{m}$ and $L = 200 \mu\text{m}$, the tunability range of $1 \delta L$, is approximatively equal to $4.78 \text{ \AA}$. So, from Fig. 12, a tuning range of $2 \delta L$ is equivalent to $\approx 9.5 \text{ \AA}$.

D. Effect of the Reflection Coefficients on the Laser Filter Spectrum

The reflection coefficient amplitudes $|r_1|$, $|r_2|$ and their phases θ_1 , θ_2 influence the laser power distribution with

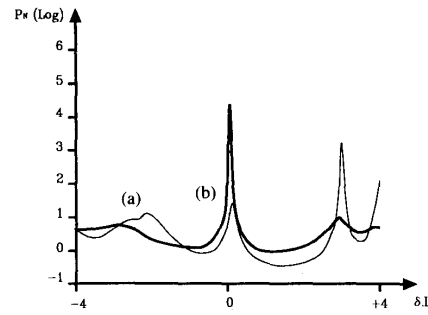


Fig. 13. Effect of end-mirror reflectivity on the emission spectra ($\alpha L = 0.1577$, $KL = 1.5$, and $\psi = \pi/2$): a) $r_1 = r_2 = 0.533$ and $\theta_1 = \theta_2 = 0$ b) $r_1 = r_2 = 0.533$ and $\theta_1 = \theta_2 = \pi/2$.

respect to the frequency [11], [13]. Fig. 13 shows two laser spectra including the effect of end mirror reflectivity.

When the two facets are cleaved ($|r_1| = |r_2| = 0.533$), the laser filter presents a multimode or a monomode spectrum operation for $\theta_1 = \theta_2 = 0$ or for $\theta_1 = \theta_2 = \pi/2$, respectively (with the phase shift ψ fixed at $\pi/2$). As shown in Fig. 13, the output spectrum shape is very sensitive to the phase of the facet reflectivity.

A single-mode operation has been also found with a cleaved front facet ($|r_1| = 0.533$, $\theta_1 = \pi/2$) and an antireflection coated end facet ($|r_2| = 0$): Fig. 14. Under these conditions, ($|r_1| = 0.533$, $\theta_1 = \pi/2$, $r_2 = 0$ and $\psi = \pi/2$), Figs. 10–12 have been plotted again and a very small difference has been found in comparison with the previous conditions ($r_1 = r_2 = 0$). Therefore it can be considered that the reflection coefficients of the laser facets have no effect on the tuning range.

E. Relation Between the Phase Shift and the Current Injected Into a Passive Region

When considering the steady state spontaneous emission of a laser, the relation between the injected current I and the carrier density N is given by

$$I = eVN(A + BN + CN^2) \quad (29)$$

where e is the electron charge, V the volume of the passive phase control wave-guide, and A , B , C coefficients relative to the different recombination processes inside the semiconductor material.

The refractive index n_{PC} and the absorption per unit length in the passive phase control region change with current injection. The relation between the refractive index change Δn_{PC} and the carrier density is given by [16]

$$\Delta n_{PC}(N) = \Gamma \frac{dn_{PC}}{dN} N \quad (30)$$

where Γ is the confinement factor and the derivative dn_{PC}/dN depends on the wavelength and the material composition of the passive layer.

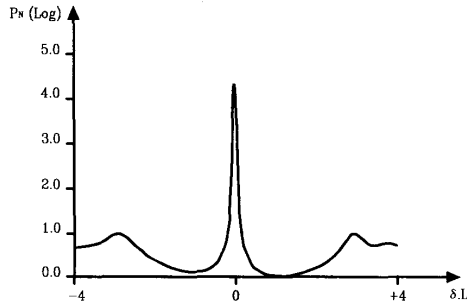


Fig. 14. Normalized emission spectra as function of δL with facet reflection $r_1 = 0.533$, $\theta_1 = \pi/2$ and $r_2 = 0$ ($\alpha L = 0.1577$, $KL = 1.5$, and $\psi = \pi/2$).

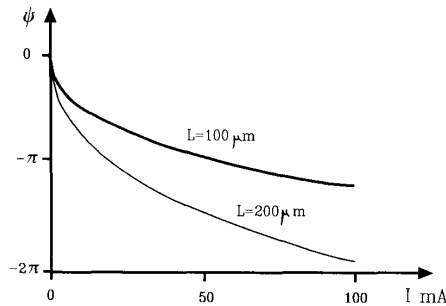


Fig. 15. Variation of the phase as function of the injected current for different values of passive region length.

The refractive index change causes a change of the wave phase $\Delta\psi$ expressed as

$$\Delta\psi(N) = 2\pi\Delta n(N) \frac{L}{\lambda} \quad (31)$$

where L is the length of the passive phase control region.

The variation of the phase as a function of the injected current is presented in Fig. 15 for different values of L with $A = 10^8 \text{ s}^{-1}$, $B = 8 \times 10^{-17} \text{ m}^3 \text{ s}^{-1}$, $C = 4 \times 10^{-41} \text{ m}^6 \text{ s}^{-1}$, and $V = 0.23 \times 1.2 \times L \text{ } \mu\text{m}^3$ [17], [18]. From Fig. 15, it can be noted that a phase shift $\Delta\psi = \pi$ is obtained when the current varies from 0 to 100 mA, and the phase control length is 100 μm .

IV. CONCLUSION

In this paper, characteristics of the spontaneous emission power spectrum of a three section phase-shift-controlled DFB semiconductor laser, biased just below the threshold condition, has been investigated using a matrix analysis. The above-developed theoretical modelization is useful for optimizing such a structure to be used as a tunable optical filter.

This study has led to the following results:

- First, the threshold conditions have been determined and the corresponding relation between the net field gain αL and the coupling coefficient κL has been theoretically developed for a structure with antireflection coated facets

and a passive region introducing a phase shift $\psi = \pi/2$. A low threshold net field gain $\alpha_{th}L$ is obtained for high coupling coefficient κL . In this case, the DFB laser filter presents a single-mode spectrum located at the Bragg wavelength when the laser operates below the threshold conditions.

- When the coupling coefficient κL gets greater, the power ratio P_{sub}/P_{main} is improved.

- The laser filter frequency depends on the phase shift ψ . When ψ is equal to $\pi/2$, the main peak power is tuned exactly at the Bragg wavelength. When $\psi > \pi/2$ or $\psi < \pi/2$, the main peak power presents a red or a blue shift tuning, respectively. The interesting feature is that the power ratio of the two peaks P_{sub}/P_{main} strongly depends on the phase shift value, especially when the lasing threshold is approached.

- According to the values of the phases θ_1 , θ_2 of the reflection coefficients r_1 , r_2 of the laser facets, the laser filter spectrum may be single-mode or multimode. But these reflection coefficients have no effect on the tuning range.

In conclusion a three section DFB laser presents a single-mode spontaneous emission spectrum operating over a wide tuning range of less than 1 nm. The results of the present analysis should aid in the design of a widely tunable optical filter, which is very interesting for high density WDM optical systems.

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REFERENCES

- [1] T. Numai *et al.*, "1.5 μm $\lambda/4$ -shifted DFB LD filter and 100 Mbit/s two-channel wavelength signal switching," *Electron. Lett.*, vol. 24, pp. 236-237, Feb. 1988.
- [2] K. Magari, H. Kawaguchi, K. Oe, Y. Nakano, and M. Fukuda, "Optical signal selection with a constant gain and a gain bandwidth by a multielectrode distributed feedback laser amplifier," *Appl. Phys. Lett.*, vol. 51, no. 24, pp. 1974-1976, Dec. 1987.
- [3] K. Magari, H. Kawaguchi, K. Oe, and M. Fukuda, "Optical narrow-band filters using optical amplification with distributed feedback," *IEEE J. Quantum Electron.*, vol. 24, pp. 2178-2190, Nov. 1988.
- [4] T. Numai, S. Murata, and I. Mito, "1.5 μm tunable wavelength filter with wide tuning range and high constant gain using a phase-controlled distributed feedback laser diode," *Appl. Phys. Lett.*, vol. 53, no. 13, pp. 1168-1169, Sept. 1988.
- [5] T. Numai, S. Murata, and I. Mito, "1.5 μm tunable wavelength filter using a phase-shift-controlled distributed feedback laser diode with a wide tuning range and a high constant gain," *Appl. Phys. Lett.*, vol. 54, no. 19, pp. 1859-1860, May 1989.
- [6] L. G. Kazovsky, M. Stern, S. G. Menocal, and C. E. Zah, "DBR active optical filters: Transfer function and noise characteristics," *J. Lightwave Technol.*, vol. 8, pp. 1441-1451, Oct. 1990.
- [7] I. M. I. Habbab, S. L. Woodward, and L. J. Cimini, "DBR-based tunable optical filter," *IEEE Photon. Technol. Lett.*, vol. 2, pp. 337-339, May 1990.
- [8] T. I. Koch, F. S. Choa, F. Heismann, and U. Koren, "Tunable multiple-quantum-well distributed-Bragg-reflector lasers as tunable narrowband receivers," *Electron. Lett.*, vol. 25, pp. 14-16, July 1989.
- [9] M. J. Chawki, R. Auffret, and L. Berthou, "1.5 Gbit/s FSK transmission system using two electrode DFB laser as a tunable FSK dis-

- criminator/photodetector," *Electron. Lett.*, vol. 26, pp. 1146-1147, July 1990.
- [10] M. J. Chawki, R. Auffret, and L. Berthou, "Multielectrode DFB lasers used as tunable FM receiver at 1.5 μm ," *J. Optical Commun.*, vol. 3, 1990.
 - [11] H. Soda and H. Imai, "Analysis of the spectrum behavior below the threshold in DFB lasers," *IEEE J. Quantum Electron.*, vol. QE-22, pp. 637-641, May 1986.
 - [12] M. Yamada and K. Sakuda, "Adjustable gain and bandwidth light amplifiers in terms of distributed-feedback structures," *J. Opt. Soc. Amer.*, vol. 4, pp. 69-76, Jan. 1987.
 - [13] J. Glinski and T. Makino, "Yield analysis of second-order DSM DFB lasers and implications for design," *IEEE J. Quantum Electron.*, vol. QE-23, pp. 849-858, June 1987.
 - [14] T. Makino and J. Glinski, "Transfer matrix analysis of the amplified spontaneous emission of DFB semiconductor laser amplifiers," *IEEE J. Quantum Electron.*, vol. 24, pp. 1507-1518, Aug. 1988.
 - [15] G. P. Agrawal and N. K. Dutta, *Long Wavelength Semiconductor Lasers*. New York: Nostrand Reinhold, 1986.
 - [16] H. Olesen, X. Pan, and B. Tromborg, "Theoretical analysis of tuning properties for a phase-tunable DFB laser," *IEEE J. Quantum Electron.*, vol. 24, pp. 2367-2375, Dec. 1988.
 - [17] A. Mozer, S. Hausser, and M. H. Pilkuhn, "Quantitative evaluation of gain and losses in quaternary lasers," in *Proc. 9th IEEE Int. Semic. Laser Conf.* Rio de Janeiro, Brazil, pp. 148-149, Aug. 1984.
 - [18] J. Manning, R. Olshansky, and C. B. Su, "The carrier-induced index change in AlGaAs and 1.3 μm InGaAs diode lasers," *IEEE J. Quantum Electron.*, vol. QE-19, pp. 1525-1530, Oct. 1983.



Mouhamad Jamil Chawki was born in Beirut, Lebanon, on June 7, 1964. He received the B.S. degree in electrical engineering (Electronic and Communication Section) from Beirut Arab University, Beirut, Lebanon in 1987, and the diplôme d'Etudes Approfondies (DEA) from the Ecole Nationale Supérieure d'Electronique et de Radio-électricité de Grenoble (ENSERG), Grenoble, France in "Optique, Optoélectronique et microondes" in 1988.

He joined the Centre National d'Etudes des Télécommunications (CNET), Lannion, France in October 1988, where he worked on the DFB laser filter applications. In March 1991, he received the "Grade de docteur de l'Université de Bretagne Occidentale en Opto-électronique." His present research interests lie in the study of multielectrode DFB and FP laser applications.



Jean Le Bihan was born in France, in 1947. From 1967 to 1971, he studied at the Ecole Normale Supérieure, Cachan, and at the Universities of Orsay and Paris. After receiving the Aggregation of Applied Physics, he was appointed to Brest Institute of Technology and, in addition, joined the Laboratory of Electronics of Brest University.

Later, he became an Assistant Professor of Electronic Engineering at Brest University and received the degree of Docteur ès Sciences Physiques for his work dealing with application of Z transform to transient analysis of distributed circuits. From 1986 to 1988, he worked as a Visiting Researcher, first in CAD of MMIC at Thomson-DAG, Corbeville, then in optoelectronics at CNET, Lannion, all in France. Since 1988, as a Professor at the Ecole Nationale d'Ingénieurs de Brest, he has been at the head of RESO Laboratory. At present, his research interests are mainly in the area of semiconductor lasers for optical communication systems.



M. Le Ligne was born in Paris, France on March 5, 1953. He received the Doctorat 3^{ème} cycle degree in signal processing from Rennes University in October 1980.

He joined the Centre National d'Etudes des Télécommunications, Lannion, France, in December 1980, where he worked on optical communication network computer simulation. Since 1988, his main research interests have been in the computer simulation of direct detection and coherent optical communication systems. He is currently studying the development of direct detection systems using optical amplifiers.



Abdelkarim Zatni was born in Casablanca, Maroc, on November 4, 1963. He received the Maîtrise of Physics in 1988 from Hassan II University, Casablanca, Maroc and the Diplôme d'Etudes Approfondies in 1989 from the University of Brest, France. He is currently working toward the Doctorate Degree at Ecole Nationale d'Ingénieurs de Brest (RESO Laboratory). His research concerns semiconductor lasers and optical communication systems.